



RENUKA KHOPALE

DATA SCIENCE INTERN | AI/ML APPLICATIONS

PROFILE

Data Science practitioner with hands-on experience in Python, SQL, Machine Learning, and Generative AI. Developed AI-based projects including a RAG-powered cold email generator using LangChain, Llama, and Streamlit, and a disease diagnosis prediction model covering the complete DL lifecycle. Skilled in data analysis, model development, and problem-solving. Previous experience as an AI and Robotics Instructor strengthened communication and technical training skills. Passionate about building AI solutions for real-world problems and continuously learning new technologies in Data Science and AI.

WORK EXPERIENCE

BRAINHAP

Robotics & Technical Instructor | October 2025 - Present (Consultant)

- Delivered Python, AI, and robotics sessions to **80+ students** using Arduino, Micro:bit, and Scratch.
- Conducted **5+ weekly interactive sessions** with a strong focus on hands-on learning and project-based activities.
- Simplified complex AI & machine learning concepts for students, strengthening communication and presentation skills.
- Created technical learning content and supported research/documentation activities, improving technical writing and collaboration abilities.

ROBOBOX

Robotics Educator | June 2025 - Sept 2025

- Trained **60+ students** in hardware programming (Arduino, Blix kits, sensors) and circuit design through a structured STEM curriculum, with **70% successfully building independent projects**.
- Applied a step-by-step problem-solving and debugging approach during training, strengthening analytical thinking and technical troubleshooting skills relevant to software development and AI/ML projects.

FELIX CONVENT

Computer Teacher | June 2024 - May 2025

- Delivered computer literacy and MS Office training to **120+ primary students** across **10 grade levels**, improving classroom task-completion confidence by **35%**.
- Designed differentiated lesson plans for 3 student ability levels, applying adaptive and analytical thinking to improve learning outcomes.

ONPRICE INFOTECH PVT. LTD.

Data Science Intern | Dec 2023 - April 2024

- Worked on data cleaning, preprocessing, and visualization using **Pandas**, **NumPy**, and **Matplotlib** for machine learning projects.
- Built and evaluated predictive machine learning models using **Scikit-learn** for classification tasks.
- Wrote **SQL** queries in **MySQL** to extract and analyze structured data for reporting purposes.
- Developed clean, modular Python code following **OOP principles** across the data pipeline.

PROJECT

AI-Powered Cold Email Generator

- Built a Generative AI app using **LangChain**, **RAG**, and **Meta Llama** for personalised cold email generation.
- Developed backend APIs & deployed an interactive **Streamlit** application.
- Reduced manual email drafting time by **75%** using automated prompt engineering and RAG.
- Implemented context-based response generation using **vector embeddings** and semantic search.

Transformative AI Technology for Disease Diagnosis

- Developed a **CNN-based** deep learning model on **12,000+ images** for plant disease detection with **94% accuracy**.
- Performed preprocessing, augmentation, training, & evaluation using **TensorFlow**, **Keras**, and **OpenCV**.
- Classified diseases using **Kaggle** and **custom agricultural datasets** with confidence-based predictions.

CONTACT

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📍 Mumbai

TECHINICAL SKILLS

Languages: Python, SQL

Data Science & ML: Pandas, NumPy, Scikit-learn, Matplotlib, TensorFlow

AI & GenAI: LangChain, RAG, Llama, Generative AI, NLP, Deep Learning

Databases: MySQL – Queries, Joins, Indexing, Data Modelling

Engineering: OOP, SDLC, REST API Integration, Streamlit Deployment

Tools: Github, VS Code, MS Office, Canva

Soft Skills: Cross-Functional Communication, Technical Instruction, Problem-Solving

EDUCATION

Master of Vocation (M.Voc)

in Software Development

Amravati University | CGPA - 8.64 | 2022

Bachelor of Science (B.Sc)

in Mathematics

Amravati University | CGPA - 6.4 | 2020

H.S.C

State Board of Maharashtra - First Class | 2017

S.S.C

State Board of Maharashtra - First Class | 2015

CERTIFICATIONS

Data Science with python | Simplilearn

Introduction to Data Science | Simplilearn